

Problem Solving with Squares and Rhombuses

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A **square** is a quadrilateral with four congruent sides and four right angles.

Rhombus

A **rhombus** is a quadrilateral with four congruent sides.

Diagonals of a Square

The **diagonals of a square** are congruent, bisect each other, and are perpendicular.

Diagonals of a Rhombus

The **diagonals of a rhombus** bisect each other at right angles and bisect the opposite angles.

Perimeter

The **perimeter** of a square or rhombus is

$$P = 4s$$

where s is the side length.

Area of a Square

The area of a square is

$$A = s^2$$

Area of a Rhombus

The **area of a rhombus** is

$$A = \frac{d_1 d_2}{2}$$

where d_1 and d_2 are the lengths of the diagonals.

Pythagorean Theorem

The diagonals of a rhombus form four right triangles, allowing the use of the **Pythagorean Theorem**.

$$a^2 + b^2 = c^2$$

Examples

Example 1

A square has a side length of 9 cm. Find its perimeter.

$$P = 4s$$

$$P = 4(9) = 36 \text{ cm}$$

Example 2

A square has a side length of 12 cm. Find its area.

$$A = s^2$$

$$A = 12^2 = 144 \text{ cm}^2$$

Example 3

The perimeter of a square is 52 cm. Find the length of one side.

$$s = \frac{P}{4}$$

$$s = \frac{52}{4} = 13 \text{ cm}$$

Example 4

The diagonal of a square is $10\sqrt{2}$ cm. Find the side length.

For a square,

$$d = s\sqrt{2}$$

Thus,

$$s = \frac{10\sqrt{2}}{\sqrt{2}} = 10 \text{ cm}$$

Example 5

A rhombus has diagonals measuring 18 cm and 24 cm. Find its area.

$$A = \frac{d_1 d_2}{2}$$

$$A = \frac{(18)(24)}{2} = 216 \text{ cm}^2$$

Example 6

A rhombus has diagonals of lengths 12 cm and 16 cm. Find the length of one side.

The half-diagonals form a right triangle.

$$s^2 = 6^2 + 8^2$$

$$s^2 = 36 + 64 = 100$$

$$s = 10 \text{ cm}$$

Example 7

In a rhombus, one diagonal measures 20 cm and the other measures 48 cm. Find the perimeter.

The half-diagonals are 10 cm and 24 cm.

$$s^2 = 10^2 + 24^2$$

$$s^2 = 676$$

$$s = 26 \text{ cm}$$

$$P = 4(26) = 104 \text{ cm}$$

Practice Exercises

Practice 1

A square has a side length of 9 cm. Find its perimeter.

Practice 2

A square has a side length of 12 cm. Find its area.

Practice 3

The perimeter of a square is 52 cm. Find the length of one side.

Practice 4

The diagonal of a square is $10\sqrt{2}$ cm. Find the side length.

Practice 5

A rhombus has diagonals measuring 18 cm and 24 cm. Find its area.

Practice 6

The diagonals of a rhombus measure 16 cm and 30 cm. Find its area.

Practice 7

A rhombus has diagonals of lengths 12 cm and 16 cm. Find the length of one side.

Practice 8

In a rhombus, one diagonal measures 20 cm and the other measures 48 cm. Find the perimeter.

Practice 9

A square-shaped garden has an area of 196 m^2 .
Find the perimeter of the garden.

Practice 10

A rhombus-shaped kite has diagonals measuring 14 cm and 22 cm. Find its area.

Activities

Activity 1

A square has a side length of 11 cm. Find its perimeter.

Activity 2

A square has a side length of 15 cm. Find its area.

Activity 3

The perimeter of a square is 68 cm. Find the length of one side.

Activity 4

The diagonal of a square is $14\sqrt{2}$ cm. Find the side length.

Activity 5

A rhombus has diagonals measuring 20 cm and 28 cm. Find its area.

Activity 6

The diagonals of a rhombus measure 18 cm and 34 cm. Find its area.

Activity 7

A rhombus has diagonals of lengths 10 cm and 24 cm. Find the length of one side.

Activity 8

In a rhombus, one diagonal measures 28 cm and the other measures 36 cm. Find the perimeter.

Activity 9

A square-shaped playground has an area of 324 m^2 . Find the perimeter of the playground.

Activity 10

A rhombus-shaped banner has diagonals measuring 18 cm and 30 cm. Find its area.

Thank you!